

DD_FA-03 - Customer KYC Field Audit

Version: v1.0 **Bundle:** 10_other **Status:** Draft for review **Primary owner:** Customer operations / compliance team **Related DDs:** DD_ILM-CFG-01, DD_FUL-02-STEP-KYC, DD_WO-01, DD_TCK-01, DD_CUST-INT-01, DD_EM-02, DD_EM-CFG-03, DD_EM-CFG-04, DD_NOT-01, DD_REP-01, FOUNDATION_FILE_STORAGE, FOUNDATION_DROOLS

MVP simplification baseline: FA-01, FA-02, and FA-03 must be implemented as one field-audit capability in the first release, with `audit_type` driving type-specific rules, checklists, and routing. Do not create three services or three independent databases.

1. Purpose

FA-03 defines field audit of customer identity, KYC evidence, contactability, address/location match, consent, and customer presence/occupancy.

Use FA-03 when:

- KYC documents submitted during sales/order need field verification;
- a customer record has duplicate, suspicious, incomplete, or expired KYC data;
- installation/support teams must confirm customer identity at site;
- compliance needs a campaign to verify customer records in an area;
- contact/address mismatch blocks activation, billing, support, or recovery;
- a rejected/incomplete KYC case needs field evidence before approval.

FA-03 does not own customer master data. ILM-CFG-01 owns customer, account, KYC, contact methods, and address records. FA-03 stores audit tasks/evidence and requests approved ILM corrections.

2. Scope And Boundaries

Capability	Owner
KYC field audit campaign, task, evidence, discrepancy	FA-03
Customer/account/KYC/contact/address source of truth	ILM-CFG-01
Order KYC step during activation	FUL-02-STEP-KYC
Field dispatch and technician execution	WO-01
Ticket/case for customer follow-up	TCK-01
Customer interaction history	CUST-INT-01
Field users/teams/contractors	EM-02
Permission model	EM-CFG-03
KYC override/correction approval	EM-CFG-04
Customer notification	NOT-01
Reporting	REP-01
Configurable KYC severity/routing	Drools

Boundary rules:

- FA-03 does not directly edit ILM customer/KYC records.
- FA-03 stores field evidence and discrepancy decisions.
- ILM-CFG-01 applies approved KYC/contact/address corrections.
- FUL-02-STEP-KYC remains the owner of order-time KYC task state.
- WO-01 owns dispatch if a field visit is required.
- KYC documents must be stored through foundation file storage with restricted access.

- EM-CFG-04 is required for risky override, KYC approval bypass, identity correction, or duplicate-merge recommendation where configured.
- Camunda is involved only when the approval policy mode in EM-CFG-04 is BPMN.

3. Process Overview

```

flowchart LR
    A["Backoffice creates KYC audit"] --> B["FA-03 loads ILM customer/KYC snapshot"]
    B --> C["Assign task or create WO"]
    C --> D["Field user verifies customer/site/evidence"]
    D --> E["FA-03 compares expected vs observed"]
    E --> F{"Issue?"}
    F -- "No" --> G["Close audit verified"]
    F -- "Yes" --> H["Open discrepancy"]
    H --> I["Drools route"]
    I --> J["ILM correction / TCK case / approval / notification"]

```

4. Status Models

Task Status	Meaning
CREATED	Audit task created.
ASSIGNED	Assigned directly or through WO.
IN_PROGRESS	Field capture started.
SUBMITTED	Observations submitted.
DISCREPANCY_OPEN	One or more issues exist.
PENDING_APPROVAL	Correction/override approval needed.
CLOSED	Audit completed.
CANCELLED	Audit cancelled.

Discrepancy Status	Meaning
OPEN	Issue detected.
ROUTED	Action selected.
PENDING_APPROVAL	Approval required.
ACTION_CREATED	ILM/TCK/NOT action created.
RESOLVED	Issue resolved.
CLOSED	Fully closed.

5. Data Model

5.1 kyc_field_audit_campaign

What it is for: stores a planned batch of customer/KYC field verification.

When to use it: create a campaign for compliance verification, duplicate-customer cleanup, rejected-KYC follow-up, or area/customer segment audit.

Sample meaning: kfac-001 verifies 500 customer KYC records in Nairobi West.

Column	Type	Purpose	Sample
campaign_id	TEXT PK	Campaign id.	kfac-001
operator_code	TEXT	Tenant/operator.	WIK
campaign_type	TEXT	KYC_REVERIFY, DUPLICATE_INVESTIGATION, ADDRESS_VERIFY, CONTACT_VERIFY.	KYC_REVERIFY
area_code	TEXT NULL	Area scope.	NRB-WEST
status	TEXT	Campaign status.	IN_PROGRESS

Column	Type	Purpose	Sample
scheduled_start_at	TIMESTAMP NULL	Planned start.	2026-08-20T08:00:00Z
scheduled_end_at	TIMESTAMP NULL	Planned end.	2026-08-30T17:00:00Z
created_by_user_id	TEXT	Creator.	usr-compliance
created_at	TIMESTAMP	Creation timestamp.	2026-08-10T09:00:00Z
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status).
- (operator_code, area_code, status).

5.2 kyc_field_audit_task

What it is for: stores one field verification assignment for a customer/account/order.

When to use it: create one task when a customer identity, address, contact method, or consent evidence needs field verification.

Sample meaning: kfat-001 asks user usr-tech-030 to verify customer CUS-100045 at the installed address.

Column	Type	Purpose	Sample
audit_task_id	TEXT PK	Task id.	kfat-001
campaign_id	TEXT NULL	Parent campaign.	kfac-001
operator_code	TEXT	Tenant/operator.	WIK
task_type	TEXT	KYC_REVERIFY, ADDRESS_VERIFY, CONTACT_VERIFY, CONSENT_VERIFY, DUPLICATE_INVESTIGATION.	ADDRESS_VERIFY
customer_id	TEXT	Customer id.	CUS-100045
account_id	TEXT NULL	Account id.	ACC-900045
order_id	TEXT NULL	Order id if order-time KYC.	ORD-1001
homepass_id	TEXT NULL	Related service location.	HP-GPON-0440
wo_id	TEXT NULL	WO if dispatch-backed.	wo-kyc-001
assigned_to_user_id	TEXT NULL	Field user.	usr-tech-030
status	TEXT	Task status.	ASSIGNED
due_at	TIMESTAMP NULL	Due timestamp.	2026-08-22T17:00:00Z
created_at	TIMESTAMP	Creation timestamp.	2026-08-20T08:00:00Z
submitted_at	TIMESTAMP NULL	Submission timestamp.	null
closed_at	TIMESTAMP NULL	Close timestamp.	null

Recommended indexes:

- (operator_code, status).
- (customer_id, status).
- (assigned_to_user_id, status).
- (wo_id).

5.3 kyc_field_audit_expected_snapshot

What it is for: stores the ILM customer/KYC/address/contact snapshot used for the audit.

When to use it: create a snapshot before field work starts, so the submitted evidence can be compared to the exact record state at assignment time.

Sample meaning: kfes-001 freezes customer name, phone, ID type, masked ID number, and expected address for CUS-100045.

Column	Type	Purpose	Sample
snapshot_id	TEXT PK	Snapshot id.	kfes-001
audit_task_id	TEXT	Parent task.	kfat-001
customer_id	TEXT	Customer id.	CUS-100045
snapshot_json	JSONB	Masked customer/KYC/contact/address snapshot.	{"name":"Jane A.", "phone":"2547***123", "kycStatus":"PENDING_REVERIFY"}
source_system	TEXT	Source owner.	ILM-CFG-01
source_version	TEXT NULL	ILM version/etag if available.	ilmv-8899
created_at	TIMESTAMP	Snapshot timestamp.	2026-08-20T08:01:00Z

Recommended indexes:

- (audit_task_id).
- (customer_id).

5.4 kyc_field_audit_observation

What it is for: stores field user observations and evidence.

When to use it: create rows for identity document sighting, customer presence, phone confirmation, address proof, site GPS, consent confirmation, or refusal.

Sample meaning: kfao-001 confirms the customer was present, phone matched, and address GPS was captured.

Column	Type	Purpose	Sample
observation_id	TEXT PK	Observation id.	kfao-001
operator_code	TEXT	Tenant/operator.	WIK
audit_task_id	TEXT	Parent task.	kfat-001
observation_type	TEXT	CUSTOMER_PRESENT, ID_SIGHTED, PHONE_CONFIRMED, ADDRESS_CONFIRMED, CONSENT_CONFIRMED, CUSTOMER_REFUSED.	ADDRESS_CONFIRMED
result_status	TEXT	PASS, FAIL, PARTIAL, NOT_VERIFIABLE.	PASS
observed_value_json	JSONB	Captured field values, masked where sensitive.	{"gpsMatch":true, "occupantConfirmed":true}
evidence_file_ids_json	JSONB	File ids.	["file-kyc-001"]
gps_latitude	DECIMAL NULL	Captured latitude.	-1.2921
gps_longitude	DECIMAL NULL	Captured longitude.	36.8219
notes	TEXT NULL	Field note.	Customer confirmed address.
captured_by_user_id	TEXT	Field user.	usr-tech-030
captured_at	TIMESTAMP	Capture timestamp.	2026-08-22T09:30:00Z
offline_client_ref	TEXT NULL	Mobile idempotency ref.	mob-kyc-888

Recommended indexes:

- (audit_task_id).
- (observation_type, result_status).
- unique (operator_code, offline_client_ref).

5.5 kyc_field_audit_discrepancy

What it is for: stores KYC/customer issues detected by field audit.

When to use it: create one row for identity mismatch, address mismatch, contact mismatch, missing consent, duplicate suspicion, refusal, or fraud risk.

Sample meaning: kfad-001 records that the installed address does not match the ILM address and requires ILM correction approval.

Column	Type	Purpose	Sample
discrepancy_id	TEXT PK	Discrepancy id.	kfad-001
audit_task_id	TEXT	Parent task.	kfat-001
discrepancy_type	TEXT	IDENTITY_MISMATCH, ADDRESS_MISMATCH, CONTACT_MISMATCH, MISSING_CONSENT, DUPLICATE_SUSPECTED, CUSTOMER_REFUSED.	ADDRESS_MISMATCH
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	Discrepancy status.	PENDING_APPROVAL
route_action	TEXT NULL	REQUEST_ILM_CORRECTION, CREATE_TICKET, REQUEST_KYC_REVIEW, REQUEST_APPROVAL, NO_ACTION.	REQUEST_ILM_CORRECTION
routed_ref_type	TEXT NULL	Target action type.	ILM_CORRECTION
routed_ref_id	TEXT NULL	Target action id.	ilm-corr-001
approval_request_id	TEXT NULL	EM-CFG-04 request.	apr-kyc-001
created_at	TIMESTAMP	Creation timestamp.	2026-08-22T09:40:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (audit_task_id, status).
- (discrepancy_type, status).
- (approval_request_id).

6. APIs

6.1 Create KYC Audit Task

POST /api/kyc-field-audits/tasks
 Idempotency-Key: kfat-CUS-100045-202608
 Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "campaignId": "kfac-001",
  "taskType": "ADDRESS_VERIFY",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "orderId": "ORD-1001",
  "homepassId": "HP-GPON-0440",
  "assignedToUserId": "usr-tech-030",
  "createWorkOrder": true
}
```

6.2 Submit Observation

POST /api/kyc-field-audits/tasks/kfat-001/observations
 Idempotency-Key: mob-kyc-888
 Content-Type: application/json

```
{
  "observationType": "ADDRESS_CONFIRMED",
  "resultStatus": "PASS",
  "observedValue": {
    "gpsMatch": true,
    "occupantConfirmed": true
  },
  "evidenceFileIds": ["file-kyc-001"],
  "gpsLatitude": -1.2921,
  "gpsLongitude": 36.8219,
  "capturedAt": "2026-08-22T09:30:00Z",
  "offlineClientRef": "mob-kyc-888"
}
```

6.3 Submit Task

```

POST /api/kyc-field-audits/tasks/kfat-001/submit
Content-Type: application/json

{
  "submittedByUserId": "usr-tech-030",
  "submittedAt": "2026-08-22T09:45:00Z"
}

```

6.4 Route Discrepancy

```

POST /api/kyc-field-audit-discrepancies/kfad-001/route
Content-Type: application/json

{
  "requestedByUserId": "usr-compliance",
  "routeAction": "REQUEST_ILM_CORRECTION"
}

```

7. Rules And Drools

Recommended packages:

- rules.field_audit.kyc.scope
- rules.field_audit.kyc.severity
- rules.field_audit.kyc.routing
- rules.field_audit.kyc.approval

```

package rules.field_audit.kyc.severity

rule "Identity mismatch is critical"
when
  $f : KycFieldAuditFact(discrepancyType == "IDENTITY_MISMATCH")
  $d : KycFieldAuditDecision()
then
  $d.setSeverity("CRITICAL");
  $d.route("REQUEST_KYC_REVIEW");
end

package rules.field_audit.kyc.routing

rule "Address mismatch requests ILM correction"
when
  $f : KycFieldAuditFact(discrepancyType == "ADDRESS_MISMATCH",
                        evidenceComplete == true)
  $d : KycFieldAuditDecision()
then
  $d.route("REQUEST_ILM_CORRECTION");
  $d.requireApproval("KYC_FIELD_AUDIT_ADDRESS_CORRECTION");
end

```

8. Events

Event	When emitted	Key fields
KycFieldAuditTaskCreated	Task created.	auditTaskId, customerId
KycFieldAuditSubmitted	Observations submitted.	auditTaskId, customerId
KycFieldAuditDiscrepancyOpened	Issue found.	discrepancyId, discrepancyType, severity
KycFieldAuditDiscrepancyRouted	Route action created.	routeAction, routedRefType, routedRefId
KycFieldAuditTaskClosed	Task closed.	auditTaskId, closedAt

9. Security And Privacy

- Do not store raw ID numbers in FA-03 tables.
- Store KYC/identity photos in restricted file storage only.
- Mask sensitive fields in snapshots.

- Field users may see only the minimum data needed to verify the task.
- Audit all reads of KYC evidence.

10. Test Scenarios

1. Task creation freezes masked ILM snapshot.
2. WO is created when dispatch is requested.
3. Mobile observation submit is idempotent.
4. Matching address/contact closes task.
5. Identity mismatch creates critical discrepancy.
6. Address correction calls EM-CFG-04 when configured.
7. ILM correction is routed through ILM API, not local DB update.
8. Raw ID number is not stored in FA tables.